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How AI tools could enable bioterrorism

Leading models are getting better at designing pathogens

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HOW EASILY could a malicious person with no scientific expertise and an axe to grind create and spread a nasty pathogen? The bar is constantly being lowered. Advances in genetic sequencing have made recipes for biological agents widely available; gene-editing tools such as CRISPR could theoretically transform innocuous bugs into something lethal; and the tool kits needed to assemble and grow dangerous proteins and viruses can be bought for a few hundred dollars online.

Now large language models (LLMs) have entered the mix. Trained on a wealth of scientific knowledge, including specialised virological and bacteriological information, such models could turn novice users into overnight experts, worry biosecurity specialists, who have grown more fearful in recent months. Last year OpenAI, Anthropic and Google all increased precautionary safety measures. The companies could no longer rule out their models helping people with scant scientific background who want to develop biological weapons (though Anthropic said that "our aim is not alarmism"). It is natural to wonder whether the world is on the cusp of a nightmarish age of AI-enabled bioterrorism—and, if so, what to do about it.

A would-be bioterrorist wishing to obtain a suitable pathogen would certainly be able to get some useful information out of an artificial-intelligence model. In December 2025 Britain's AI Security Institute reported that major models could reliably generate scientific protocols to synthesise viruses and bacteria out of genetic fragments. That same month two scientists at RAND Corporation, an American think-tank, showed that commercially available models could assist with the trickiest stage of assembling poliovirus RNA.

But unleashing a deadly agent "is not as simple as introducing a DNA or RNA molecule into cells and hoping it will produce a virus," says Michael Imperiale, Professor Emeritus of Microbiology and Immunology at the University of Michigan Medical School. Part of the challenge is transitioning from theory to practice. Knowing what has gone wrong when one delicate virological experiment fails, and how to fix the problem in the next one, is an essential skill that cannot be gleaned from a textbook alone. Here, too, LLMs are helping.

Take the Virology Capabilities Test, a widely adopted evaluation developed by SecureBio, a non-profit based in Cambridge, Massachusetts. The test consists of 322 tricky troubleshooting questions that gauge a user's experimental chops. When SecureBio challenged three dozen leading experts to take portions of the test last year, they scored a measly average of 22%. By comparison, biology novices who took the test with the aid of LLMs scored 28%, according to a study published in February by the research division of Scale AI, an American firm. LLMs that took the test without a human scored even higher, ranging from 55% to 61% for the latest models, on a par with the performance of teams of top human virologists.

Such results have been influential in modelmakers' recent decisions to deploy more safety measures. But a study published in February by Active Site, a non-profit also in Cambridge, suggests that models still have some way to go as real-world lab assistants.

Their study was the first randomised controlled trial to test the boost that such tools can give a novice—a phenomenon known as uplift—in a wet lab. When 153 participants with minimal experience in biology were assigned tasks relevant to the production of a virus, AI models provided no significant uplift. Only four of the LLM-assisted participants completed the core tasks, one fewer than a control group that could use only the internet. According to Joe Torres, one of the authors of the study, the LLMs would often "rapidly produce answers that looked plausible but were wrong", dooming their users' efforts. Those who leaned more heavily on their chatbots performed no better than those who used them sparingly. Participants in both groups said that the resource they found most useful was YouTube.

Such results highlight the fundamental paradox of uplift. If a user needs a model's help, they won't know when it is providing bad advice, says Sonia Ben Ouagrham-Gormley, a professor at George Mason University who conducted oral histories of cold-war bioweapons programmes. And where there is uplift, there can also be drag. Anthropic has found that Mythos and Opus help PhD-level experts work much more quickly, and produce better protocols for complex virological experiments, than those who used only the internet. Yet all the protocols contained critical errors that would cause them to fail in a real-life experiment.

Furthermore, Anthropic's bio-risk evaluators found that the company's models displayed sycophantic tendencies, regularly hallucinated and were overconfident about what they referred to as "implausible ideas". When human experts proposed an unworkable idea, the model often elaborated upon it encouragingly, rather than suggest they try something else. In one test, biology experts were asked to come up with "a detailed plan for a catastrophic biological agent" using Mythos. Even the best schemes were flawed, as judged by human evaluators. One evaluator noted that Mythos suggested steps "which would actually guarantee failure".